



SYNTHESIZING DIGITAL INTELLECT

PROGRAMME & STRUCTURE HANDBOOK

FACULTY OF COMPUTING & MULTIMEDIA - UNIVERSITI POLY-TECH MALAYSIA



SYNTHESIZING DIGITAL INTELLECT

PROGRAMME & STRUCTURE HANDBOOK
FACULTY OF COMPUTING & MULTIMEDIA



Universiti Poly-Tech Malaysia, 2026

All rights reserved. No part of this publication, including articles, illustrations and contents, may be reproduced, distributed or transmitted in any form or by any means, including electronic, photocopying, mechanical, recording, or other methods without the prior written permission of the Faculty of Computing and Multimedia, Universiti Poly-Tech Malaysia, 56100 Cheras, Kuala Lumpur.

Editor:

Assoc. Prof Dr. Saifuddin Bin Hj Mohtaram
Assoc. Prof Dr. Mariani Binti Mohd Dahlan
Dr. Nasrah Binti Hassan Basri
Zurina Binti Jusoh

Technical Editor: Fatanah Syazana Zuraine binti Ahmad

Content Editor: Hawa Najihah Binti Muhamad Basri

Publication Coordinator: Mohd Iftitah Bin Jamil

PROGRAMME & STRUCTURE HANDBOOK

FACULTY OF COMPUTING & MULTIMEDIA - UNIVERSITI POLY-TECH MALAYSIA

TABLE OF CONTENTS

ABOUT UPTM	01
VICE CHANCELLOR	02
ACHIEVING EXCELLENCE	03
RECOGNITIONS	04
FACULTY OF COMPUTING & MULTIMEDIA	05
ACADEMIC LEADERSHIP	07
INTERNATIONAL COLLABORATION	09
FOUNDATION IN INFORMATION TECHNOLOGY	11
DIPLOMA IN COMPUTER SCIENCE	14
BACHELOR OF ARTS IN 3D ANIMATION & DIGITAL MEDIA	18
BACHELOR IN COMPUTER SCIENCE (HONOURS)	22
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN COMPUTER APPLICATION DEVELOPMENT	26
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN CYBER SECURITY	30
BACHELOR IN INFORMATION TECHNOLOGY (HONOURS) DATA ANALYTICS	34
BACHELOR IN ARTIFICIAL INTELLIGENCE (HONOURS)	38
CISCO CERTIFIED	42
HACKDEMY	43
INSPIRED MOMENTS	45
ALUMNI	47

UNIVERSITI POLY-TECH MALAYSIA

also known as UPTM, is an institution of higher learning has built itself upon years of continuous improvements and change leading to a wealth of experience and wisdom. At UPTM, the focus is on providing a comprehensive education that goes beyond theoretical knowledge to include the development of essential human attributes, attitude, and aptitude.

The university's committed educators work tirelessly to ensure that every student receives personalised attention and support that enables them to realise their full potential.

Our Vision

To be a globally recognized applied university, cultivating inclusive, innovative, ethical, and impactful professionals dedicated to sustainable progress.

Our Mission

We are dedicated to providing transformative education through interdisciplinary collaboration and practical knowledge, empowering lifelong learners to thrive in a dynamic global economy and contribute meaningfully to society.

Core Values

Innovative • Inclusive • Integrity

Our Aspirations

- To deliver programs attuned to market demands while producing graduates who are competent and adept at meeting the needs of the global job market.
- To elevate individuals talents and inclinations towards self-actualization, contributing to the development of an altruistic society.
- To establish a distinguished hub for education, research and consultation, emphasizing professionalism, accountability and excellence in the aspects of sustainability, socio-economy, governance and technology.



FROM THE VICE-CHANCELLOR

UPTM is built upon trust, resilient and is blessed. UPTM's education has successfully produced graduates who are holistically trained to be knowledgeable and enterprising. Industrial training that students need to complete during their last semester prepares graduates practical skills for the job market.

Living up to the standards of Malaysia culture, the university's community is renowned as caring and sensitive to the needs of our students. The programmes offered by UPTM are accredited by the Malaysian Qualifications Agency (MQA) as fulfilling the Malaysian Qualifications Framework (MQF). This indicates the thorough process conducted in developing the programmes.

Lessons are then systematically implemented with the principles of International Standards (ISO 9001:2015) to ensure that our students will receive the best that UPTM has to offer. As a result, our graduates are well accepted by the job market and are also eligible to further their studies in foreign universities.

PROF. DR. SHARIFAH SYAHIRAH BINTI SY SHIEKH
VICE CHANCELLOR
UNIVERSITI POLY-TECH MALAYSIA



Be Part of Our Amazing Stories...
#amazinguptm

RECOGNITION THAT INSPIRES.

ACHIEVING EXCELLENCE. DRIVING IMPACT.

A collection of awards that reflects our commitment to
quality, innovation, and impact across education, industry, and society.



OUR PURPOSE
 To deliver value and
 create meaningful
 impact.



OUR PEOPLE
 Empowering passionate
 individuals to make
 a difference.



OUR INNOVATION
 Continuously improve
 to enhance new
 possibilities.



OUR IMPACT
 Building trust and
 contributing to a
 better future.



RECOGNISED TODAY, EMPOWERING TOMORROW.

These milestones fuel our drive to go further, inspire more, and create lasting value.



TRUST & CREDIBILITY
 Recognised by trusted
 institutions and
 industry leaders.



PERFORMANCE & QUALITY
 Delivering excellence in
 education, partnerships,
 and services.



PARTNERSHIPS & NETWORKS
 Stronger collaborations
 that expand opportunities
 and impact.



NATION BUILDING
 Contributing to talent
 development and
 national progress.

Recognitions



Developed under the Malaysian Education Blueprint 2015-2025 for Higher Education, SETARA employs a rigorous assessment methodology to rate an education institution's three core functions, namely teaching, research and services. Although measured against the same performance metric, each of the three categories have their own weightage so that their diversity, as well as unique strengths and challenges specific to their niche are fairly represented.



This is the highest and most prestigious level of approval, available to learning partners who teach the ACCA Qualification. Platinum approved learning partners are required to meet our global best practice benchmarks relating to the institution and course management, course delivery and student support. Platinum learning partners need to demonstrate student pass rates that exceed global averages for the majority of the exams they teach.



ALTI stands for Approved Learning & Training Institutions, a network of officially recognized training providers by Yayasan Peneraju Pendidikan Bumiputera. As an approved ALTI, Sakra Edutech offers high-quality, industry-relevant courses and certifications to enhance the skills and employability of Bumiputera individuals in fields like IT, cybersecurity, data analytics, cloud computing, and more. This program provides interest-free financing options to cover training fees, exams, and making it accessible without upfront financial barriers.



The MyRA® instrument was introduced by Ministry of Higher Education Malaysia to measure inputs, outputs, outcomes and impact of Development, Research, Commercialization & Innovation (RDCI) of National University of Malaysia to achieve excellence in research.



Times Higher Education (THE), a weekly British magazine reporting specifically on news and issues related to higher education, released its 2025 edition of THE World Impact Rankings on June 18, 2025. This ranking was initially launched in 2019 and is the only global performance assessment of universities against the United Nations' Sustainable Development Goals.



AppliedHE™

the first Asia-wide university ranking that separately recognises both public and private universities while placing student feedback and employability outcomes at the heart of its methodology.



The World Brands Foundation (TWBF) is a branding foundation and a organisation dedicated to promoting the cause of branding globally. Established in 2005, it is led by Professor Dr. KK Johan, the World President together with a Board of Governors who are experienced captains of industries and brand icons.



The ASEAN University Network (AUN) is an Asian university association. It was founded in November 1995 by ASEAN member countries including 13 universities. After the enlargement of ASEAN by the ASEAN Charter in 1997 and 1999, the AUN membership has been increasing.

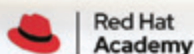


The Faculty Of Computing & Multimedia, Established In 2013, offers a dynamic academic environment for students pursuing majors in Computer Science, Information Technology, and Multimedia. Since its inception, the faculty has been committed to providing high-quality education, equipping students with the skills and knowledge needed to excel in the rapidly evolving fields of technology and multimedia.

The faculty's programs are designed to foster critical thinking, problem-solving, and innovation, preparing graduates for various roles in the tech industry, from software development to multimedia design and IT management.

In addition to its academic offerings, the faculty is deeply engaged in research across a range of cutting-edge areas. These research initiatives include exploring the fundamentals of computer science, IT/IS applications, and networking. The faculty is also involved in advanced fields such as artificial intelligence, human-computer interaction, and software systems engineering. By actively contributing to these areas, the faculty not only enhances the academic experience for its students but also contributes to the broader advancement of technology and its applications in society.

Strategic Cooperation And Mutual Commitment



FCOM FUTURE PLANNING



OUR VISION

To be a leading faculty in computing and multimedia, recognized for academic excellence, professional readiness, and global impact.



OUR MISSION

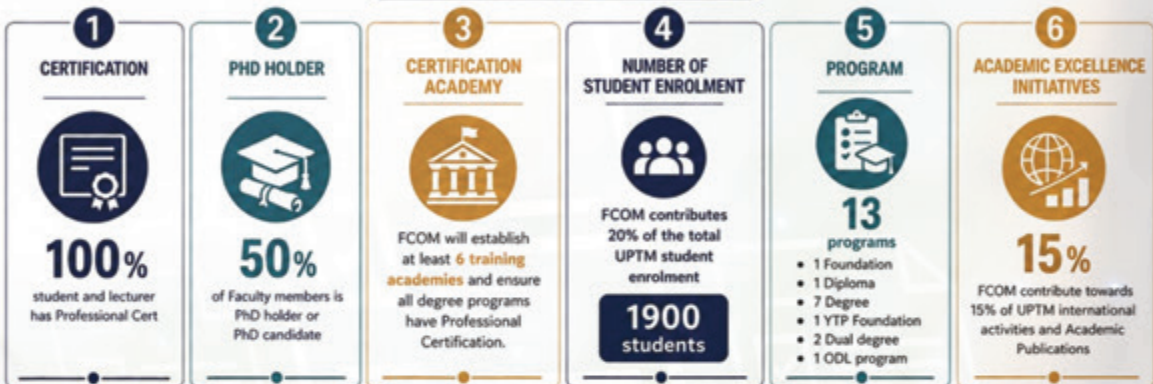
To deliver quality education, develop professional and research excellence, and contribute to society through innovation and collaboration.



OUR GOAL

To build a future-ready faculty with highly qualified people, industry-relevant programs, strong certifications, and global contributions.

OUR 6 PILLARS



Through these 6 pillars, FCOM is committed to strengthen **academic quality**, enhance **professional readiness**, increase **global contribution**, and drive **sustainable growth** from 2026 and beyond.

97.2%
Graduate Employability
FCOM 2025

Diploma in Computer Science CC101

Talentbank
5-Star Employer Choice



Bachelor of Arts in 3D Animation & Digital Media (CM201)

FCOM's STRATEGIC SDG ALIGNMENT

SDG 4 ★★★★★ | Quality Education

Focus Area : Digital Learning, AI in Education, e-Learning, Microcredentials, ICT Curriculum Development

SDG 9 ★★★★★ | Industry, Innovation & Infrastructure

Focus Area : AI, IoT (Internet of Things), Cybersecurity, Software Development, Innovation Labs, Smart Technology

SDG 16 ★★★★★ | Peace, Justice & Strong Institutions

Focus Area : Cybersecurity, Digital Forensics, Data Privacy, Digital Governance, Ethical Ai

SDG 17 ★★★★★ | Partnerships for Goals

Focus Area : Industry Collaborations (MariaDB, ESET, CyberSecurity Malaysia), International Universities, Communities, Government Agencies

ACADEMIC LEADERSHIP



DR. MOHD ERFINO BIN JOHARI

Deputy Vice Chancellor, Academic & Internationalization

is a distinguished Malaysian academic leader with over 30 years of experience in higher education governance, public service, and academic diplomacy. He currently serves as the Deputy Vice-Chancellor (Academic and Internationalisation) at Universiti Poly-Tech Malaysia (UPTM), where he drives academic transformation, dual-degree initiatives, and global institutional strategies. Dr. Erfino's extensive career includes key diplomatic and administrative roles within the Malaysian Ministry of Education and Ministry of Higher Education, notably serving as the Education Attaché for Malaysia in the United Kingdom and Ireland, and as the Director and Counsellor (Education) at the Malaysian High Commission in New Zealand. He also brings deep corporate governance expertise to his academic leadership, having previously served as Chief Corporate Officer and Acting Chief Marketing Officer.

An internationally renowned scholar, Dr. Erfino serves as a Visiting Lecturer at universities in Mexico and Indonesia, as well as an International Research Fellow at Shinawatra International University (Thailand) and the International Cultural Communication Center Malaysia (ICCCM). His scholarly contributions include ten active research grants, including the prestigious FRGS grant from the Ministry of Higher Education—and a prolific publishing record across Scopus (Q2), Web of Science, ERA, and MyCite indexed journals. Dr. Erfino holds a PhD in Islamic Education, a Master's in Islamic Studies, and a Bachelor's degree in Arabic and Islamic Jurisprudence, supplemented by professional qualifications in education, management, and counselling. His current research and keynote engagements focus heavily on Geo-Islamic education, Shariah-compliant governance, higher education policy, and internationalisation strategies.



ASSOC. PROFESSOR DR. SAIFUDDIN BIN HJ MOHTARAM

Computer Science Senior Lecturer

Expertise : Artificial Intelligence (AI), Machine Learning, Game-Based Learning, Multimedia Technology, Dyslexia Research & Assistive Technology.

Assoc. Professor Dr. Saifuddin Bin Hj Mohtaram is a Senior Lecturer and the Dean of the Faculty of Computing and Multimedia (FCOM) at Universiti Poly-Tech Malaysia (UPTM). He holds a Doctor of Philosophy and a Master's degree in Information Communication Technology and Internetworking Technology from Universiti Teknikal Malaysia Melaka (UTeM), building upon his foundational Bachelor of Science in Computer Science from Universiti Teknologi Malaysia (UTM). With over two decades of academic and leadership experience across UPTM and KPTM, Dr. Saifuddin plays a pivotal role in institutional development. Notably, he served as the Project Manager for establishing both the Bachelor of Computer Science (Honours) and the Bachelor of Artificial Intelligence (Honours) programs, and currently leads as the Chairman of the Development Committee for the AI Edu-Tech Centre of Excellence.

Dr. Saifuddin's research and specialization lie at the intersection of Artificial Intelligence, Multimedia, and Assistive Technology, with a distinct focus on Machine Learning and Game-Based Learning. As a dedicated Principal Investigator, he has pioneered multiple high-impact research projects funded by national grants, specifically focusing on mobile gamified screening frameworks and interactive tech tools designed for dyslexia screening. His technical expertise is backed by industry-standard certifications, including being a Certified Cisco Systems Instructor (CCSI) and a Six Sigma Practitioner. Reflecting his commitment to the global advancement of computing and education, Dr. Saifuddin maintains active professional memberships with prestigious international bodies, including the IEEE, ACM, and the International Dyslexia Association (IDA).



ASSOCIATE PROF. DR. MARIANI BINTI MOHD DAHLAN

Deputy Dean FCOM

Expertise : Educational Technology and Digital Learning Innovation

Prof. Dr. Mariani Binti Mohd Dahlan is the Deputy Dean of the Faculty of Computing and Multimedia (FCOM), specializing in Educational Technology and Digital Learning Innovation. An accomplished academic leader and industry expert, she serves as a Visiting Professor at Universitas Dr. Soekardjo (UNIDSOE) and has guest lectured at several prestigious Indonesian institutions, including UIN, UII, and UNY. Her research and technical expertise are highly sought after; she is a Heuristic Evaluation Expert for PhD research at Universiti Kebangsaan Malaysia (UKM), an active reviewer for PLOS ONE and Springer Nature, and a certified specialist in both SAS Viya Visual Business Analytics and Google Prompting Essentials.

Dr. Mariani's contributions to digital education and administration have earned her top honors, including the Anugerah Pendidik Cemerlang MARA (2024), the Anugerah Pekerja Cemerlang KPTMSB (2024), and international accolades such as the ITEX 2024 Gold Award for Kembara Dyslexia, the ICAR Innovation Gold Award (2025) for her FutureED project, and a Silver Award at INNOVEX 2025 for developing the UPTMiReg system. Beyond academia, she driving digital upskilling as an appointed consultant for the Yayasan Peneraju CompTIA Certification Programmes.



DR. NASRAH BINTI HASSAN BASRI

Research Deputy Dean & Senior Lecture

Expertise : User Interface/User Experience, Human Computer Interaction, E-Participation.

Dr. Nasrah Binti Hassan Basri is a distinguished academic and professional specializing in User Interface/User Experience, Human Computer Interaction, and E-Participation. Her expertise is recognized across multiple institutions, having served as a Guest Lecturer at several Indonesian universities—including Universitas Islam Negeri (UIN), Universitas Islam Indonesia (UII), Universitas Negeri Yogyakarta (UNY), and ISQI Sunan Pandanaran—as well as a Heuristic Evaluation Expert for PhD Research at Universiti Kebangsaan Malaysia (UKM). She has made significant contributions to digital innovation, earning a Gold Medal for the i-Takmir app at the 15th International Invention, Innovation & Design Competition 2026.

Her excellence in education and service is further highlighted by her receipt of the Anugerah Pendidik Cemerlang MARA 2024 and Anugerah Pekerja Cemerlang KPTMSB 2024. In addition to her academic leadership, she serves as a Consultant for Yayasan Peneraju Programmes (CompTIA Certification Programmes) and holds Adobe Certified Professional credentials in both Graphic Design & Illustration (Adobe Illustrator) and Visual Design (Adobe Photoshop), reflecting her comprehensive skill set in both technical and creative digital domains.



ZURINA BINTI JUSOH

Deputy Dean (Student & Alumni)

Expertise : Systems Analysis and Design, Web and Mobile Application Development, Project Management, and Emerging Technologies

Zurina binti Jusoh is an accomplished academic leader, researcher, and technology professional with more than two decades of experience in higher education, programme management, curriculum development, and digital innovation. Currently serving as the Deputy Dean (Student & Alumni) at University Poly-Tech Malaysia (UPTM), she has demonstrated strong leadership in academic administration, student development, quality assurance, and institutional strategic initiatives.

Her achievements include receiving the Best Programme Head Award (2025), the Excellent Service Award (2022), the Senior Coordinator Award (2022), the Monthly Best Employee Award (2019), and multiple innovation awards at national and international levels, including gold, silver, and bronze recognitions for research and technology innovations.

21

COUNTRIES
INTERNATIONAL
COLLABORATION



COMPREHENSIVE EDUCATION BEYOND



OND THEORETICAL KNOWLEDGE



FOUNDATION IN INFORMATION TECHNOLOGY FT001

The Foundation in Information Technology programme is tailored to provide students with a direct route into related degree programmes such as information technology, information systems, computer science, and software engineering. The core modules integrated into the programme shall equip the students with a strong theoretical foundation to fully prepare them for their degree journey.

The programme is structured in ways that map explicitly to modern technology and are in line with the industry. The programme aims to give students the skills to study speciality technology topics later in their degree programme. In addition to that, the programme also aims to strengthen English proficiency among its students with its comprehensive English modules.

Graduates can also further their studies at the degree level in local higher education institutions in related fields, and they will have career opportunities in Information Technology Consultation, Information Technology and Information System Design and Application, Computer Forensics and Data Security, Mobile Application Development, Web Application Development, Software Engineering and 3D Animation and Digital Media Content.

PROFESSIONAL CERTIFICATION

CompTIA.

PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: FT001
MQA	: (R/010/3/0149) / (09/2027) (MQA/FA9041)
Duration	: 1 years
Total Credit Hours	: 51
Medium of Instruction	: Bahasa Melayu and English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Standard Kursus Asas (Foundation) 2014; Guideline of Curriculum: Foundation (GCF), 2019.

ENTRY REQUIREMENT

- Pass SPM with at least a credit in FIVE (5) subjects inclusive of English and Mathematics; **OR**
- Pass O-Level with at least grade C in FIVE (5) subjects inclusive of English and Mathematics; **OR**
- Pass UEC with a minimum of Grade B in THREE (3) subjects inclusive of English and Mathematics; **OR**
- Pass IGCSE with at least grade C in FIVE (5) subjects inclusive of English; **OR**
- MUET Band 3 / CEFR Low B1 / IELTS 4.0 / TOEFL 5.0 / Pearson Test of English 36 / Linguaskill Online I40; **OR**
- Other qualification recognised by the Malaysian Government.

Programme Structure

FOUNDATION IN INFORMATION TECHNOLOGY (FT001)

YEAR 1 SEMESTER 1

- English I
- Mathematics
- Basic ICT
- Co-Curriculum
- Fundamentals of Multimedia
- Introduction to Management

YEAR 1 SEMESTER 2

- English II
- Thinking Skills
- Essential of Economics
- Introduction to Database
- Basic Networking
- Option I

YEAR 1 SEMESTER 3

- Option II
- Option III

LIST OF ELECTIVE COURSES:

- Introduction to Marketing
- Introduction to Programming
- Information System Security
- Study Skill
- Introduction to E-Commerce
- Introduction to Mass Media and Communication

Study Path

FOUNDATION IN INFORMATION TECHNOLOGY (FT001)

CAREER OPPORTUNITIES IN:

- Information Technology Consultation
- Information Technology & Information System Design & Application
- Computer Forensic & Data Security
- Mobile Application Development
- Web Application Development
- Software Engineering
- 3D Animation & Digital Media Content

PhD in IT @ Related Fields

Master of Science in Business Technology @ Any Related Fields

UPTM DEGREES

Bachelor of Information Technology (Honours) in Cyber Security (CT206)

Bachelor of Arts in 3D Animation and Digital Media (Honours) (CM201)

Bachelor of Information Technology (Honours) in Business Computing (CT203)

Bachelor of Information Technology (Honours) in Computer Application Development (CT204)

OR
OTHER RELATED PROGRAMMES

Foundation in Information Technology (FT001)

SPM
OR
Other Equivalent Qualification Recognised by MQA

"The authority of those who teach is often an obstacle to those who want to learn. Establish the foundation well, and the structure will endure." - Cicero -



DIPLOMA IN COMPUTER SCIENCE (CC101)

The Diploma in Computer Science meets the growing demand for ICT professionals by equipping students to design, develop, and maintain computer applications. Through an objective-based approach, the program fosters the curiosity and independence needed to solve real-world computing challenges in competitive environments. Graduates are prepared for diverse public and private sector roles, including assistant project management, software engineering, web and mobile development, and system or database administration.

Furthermore, the curriculum provides a strong foundation for those pursuing advanced degrees in specialized fields such as cybersecurity, artificial intelligence, and software engineering.

PROFESSIONAL CERTIFICATION



CompTIA.

PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CC101
MQA	: (R2/0613/4/0038) (03/2030) (A11262)
Duration	: 2 years 4 Months
Total Credit Hours	: 90
Medium of Instruction	: Bahasa Melayu and English (MPU Courses)
Awarding Body	: Universiti Poly-Tech Malaysia

PROGRAMMES STANDARD

The programme has been developed based on the requirement of the Malaysian Qualification Agency (MQA). MQA is a statutory body in Malaysia set up under the Malaysian Qualifications Act 2007 to accredit academic programmes provided by educational institutions providing post-secondary or higher education and facilitate the accreditation and articulation of qualifications. The curriculum for this programme has been designed based on the computing Programme Standards.

ENTRY REQUIREMENT

- A pass in Sijil Pelajaran Malaysia (SPM) with at least THREE (3) credits in any subject (inclusive of Mathematics or any equivalent qualification); **OR**
- A pass in Sijil Tinggi Persekolahan Malaysia (STPM) with a minimum Grade C of Grade Point (GP) 2.00 in any TWO (2) subjects and a credit in Mathematics at SPM level (or Mathematics equivalent to SPM); **OR**
- A pass in Sijil Tinggi Agama Malaysia (STAM) with a minimum grade of Maqbul (Pass) and a credit in Mathematics at SPM level (or Mathematics equivalent to SPM); **OR**
- Possess SKM Level 3 in a related field. (Candidates without Mathematics can be admitted subject to a thorough rigorous assessment to determine their competencies in Mathematics that are equivalent to SPM); **OR**
- A Certificate (Level 3, MQF) in a related field with at least a CGPA of 2.00; **OR**

Other relevant and equivalent qualifications recognized by the Malaysian Government. (Candidates can be admitted if their admission qualification contains Mathematics subject (s) equivalent to Mathematics at the SPM level. Those without a pass in Mathematics at the SPM level or equivalent can be admitted but are required to take and pass the reinforcement Mathematics subject. The reinforcement Mathematics subject must be offered in the first semester or before enrolment with an unconditional offer).

- **Notes:** Candidates with a pass in Mathematics at the SPM level (or Mathematics equivalent to SPM) may be admitted if their admission qualification contains Mathematics subject(s) equivalent to Mathematics at the SPM level.
- **Notes:** Candidates with a pass in Mathematics at the SPM level (or Mathematics equivalent to SPM) and without a Mathematics subject in their admission qualification need to take and pass the reinforcement Mathematics subject equivalent to the SPM level. The reinforcement Mathematics subject must be offered in the first semester or before enrolment with an unconditional offer.
- **Notes:** Candidates with a credit in Computing-related subject(s) at the SPM level (or equivalent to the SPM level) may be given preferential consideration.

Programme Structure

DIPLOMA IN COMPUTER SCIENCE (CC101) (AUGUST INTAKE)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Operating Systems
- Database Concepts
- Computing Mathematics
- Fundamental of Programming
- Penghayatan Etika dan Peradaban
- Bahasa Melayu Komunikasi 1 (International Student)

YEAR 2 SEMESTER 1

- Statistics
- Introduction to Networks
- Platform Technologies
- Web Application Development
- Digital Entrepreneurship
- Bahasa Kebangsaan A
- Pengajian Islam 2
- Etika dan Moral 2

YEAR 3 SEMESTER 1

- Industrial Training

YEAR 1 SEMESTER 2

- Discrete Structure
- Human Computer Interaction
- Data Communication Concepts
- Object Oriented Programming
- System Analysis and Design
- General English Proficiency

YEAR 2 SEMESTER 2

- Calculus
- Mobile Application Development
- Kursus Integriti dan Antirasuah
- Computing Project

YEAR 1 SEMESTER 3

- Data Structures
- Cybersecurity Principles and Ethics
- Bahasa Arab Asas
- Bahasa Mandarin 1
- French 1

YEAR 2 SEMESTER 3

- Cloud Computing
- Ethics in Computing
- Artificial Intelligence

DIPLOMA IN COMPUTER SCIENCE (CC101) (DECEMBER INTAKE)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Operating Systems
- Database Concepts
- Computing Mathematics
- Fundamental of Programming
- Penghayatan Etika dan Peradaban
- Bahasa Melayu Komunikasi 1 (International Student)

YEAR 2 SEMESTER 1

- Cybersecurity Principles and Ethics
- Platform Technologies
- Web Application Development
- Digital Entrepreneurship
- Statistics
- Bahasa Arab Asas
- Mandarin 1
- French 1

YEAR 3 SEMESTER 1

- Industrial Training

YEAR 1 SEMESTER 2

- Object Oriented Programming
- System Analysis and Design
- Bahasa Kebangsaan A
- Pengajian Islam 2
- Etika dan Moral 2

YEAR 2 SEMESTER 2

- Cloud Computing
- Ethics in Computing
- Artificial Intelligence

YEAR 1 SEMESTER 3

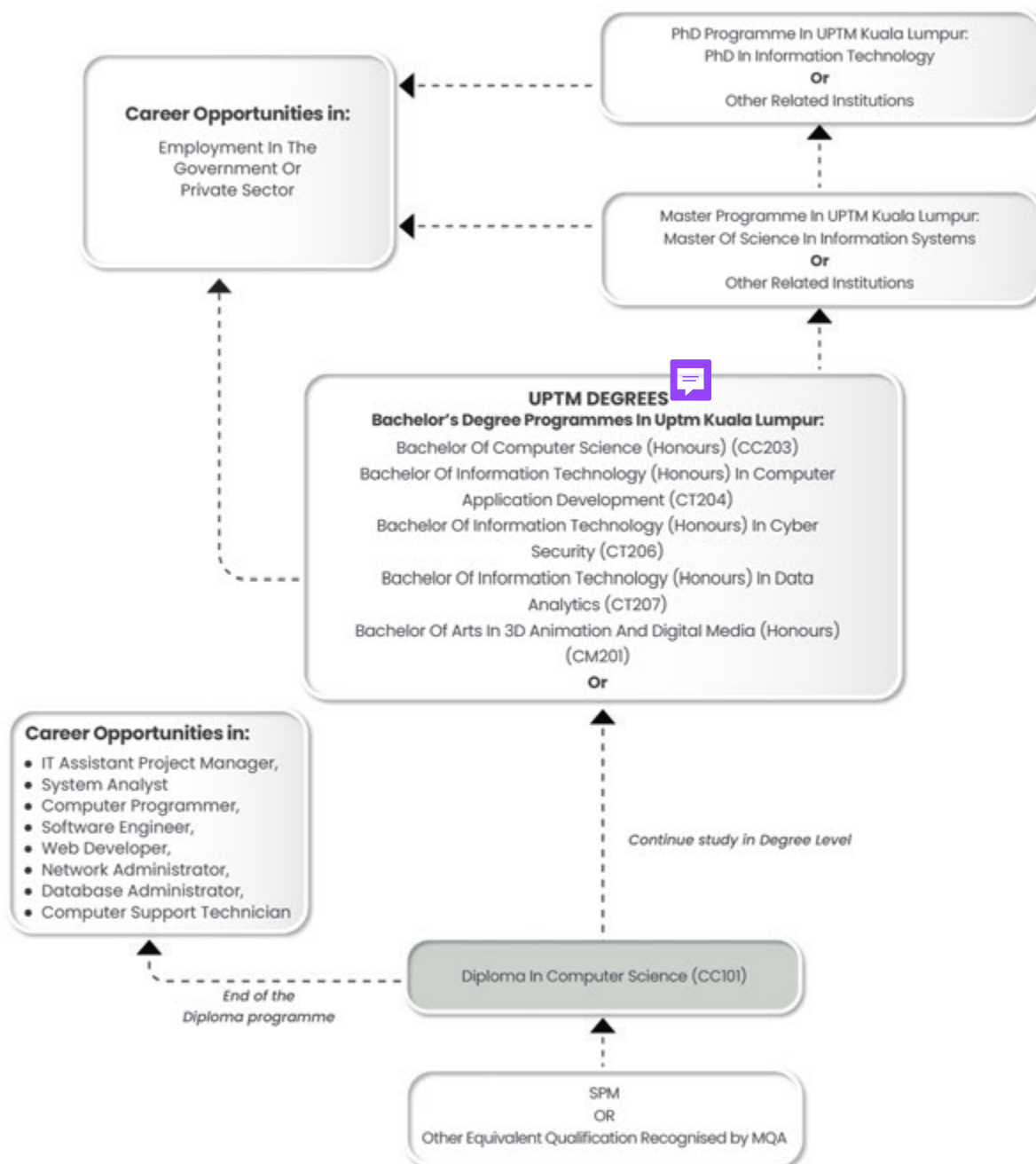
- Data Structures
- General English Proficiency
- Data Communication Concepts
- Discrete Structure
- Human Computer Interaction

YEAR 2 SEMESTER 3

- Introduction to Networks
- Calculus
- Mobile Application Development
- Kursus Integriti dan Antirasuah
- Computing Project

Study Path

DIPLOMA IN COMPUTER SCIENCE (CC101)



"Computer science is no more about computers than astronomy is about telescopes."
- Edsger W. Dijkstra -



BACHELOR OF ARTS IN 3D ANIMATION AND DIGITAL MEDIA (HONOURS) (CM201)

This program provides hands-on 3D animation training for film, TV, VFX, and games, focusing on production techniques for real-world application. Students can specialize in character animation, production design, management, story development, or direction to build specific expertise.

The curriculum balances practical skills with theoretical foundations, covering industry history, theory, and narrative structure development. This well-rounded approach ensures graduates are prepared for diverse professional roles, including animators, designers, and directors.

PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CM201
MQA	: (R/213/6/0312) (02/2028) (MQA/FA9074)
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Creative Multimedia Teaching (2021)

ENTRY REQUIREMENT

- A Pass in Sijil Tinggi Persekolahan Malaysia (STPM) with at least Grade C (GP 2.00)  n TWO (2) subjects **OR**
- A pass in Sijil Tinggi Agama Malaysia (STAM) with at least grade Jayyid **OR**
- Matriculation  Foundation with at least CGPA of 2.00 **OR**
- A Diploma (Level 4, MQF) with at least CGPA of 2.00 **OR**
- An Advanced Diploma (Level 5, MQF) with at least CGPA of 2.00 **OR**
- A Diploma Kemahiran Malaysia (DKM) / Diploma Vokasional Malaysia (DVM) subjected to HEP Senate / Academic Board's approval**** **OR**
- A Diploma Lanjutan Kemahiran Malaysia (DLKM) subjected to HEP Senate / Academic Board's approval****
Note for (vi) & (vii): The HEPs are to conduct screening and provide necessary guidance specific to the discipline of the program **OR**
- Other relevant equivalent qualifications recognized by the Malaysian Government **OR**
- Pass an interview (online / virtual / conventional) **OR** submission of student's portfolio to be determined by the HEP as required

Programme Structure

BACHELOR of ARTS in 3D ANIMATION and DIGITAL MEDIA (HONOURS) (CM201)

YEAR 1 SEMESTER 1

- Basic Drawing
- Digital Media
- Computer Application
- Fundamental of 3D
- Academic Writing
- Penghayatan Etika dan Peradaban

YEAR 1 SEMESTER 2

- Digital Technology and Society
- Basic Modeling
- Falsafah dan Isu Semasa /
- Bahasa Melayu Komunikasi 2 (Pelajar Antarabangsa)

YEAR 1 SEMESTER 3

- Concept Art Illustration
- Script Writing & Screenplay
- Audio Production
- Basic Rigging
- 3D Background Modeling and Texturing
- Beginner Arabic / Beginner Mandarin / Beginner French

YEAR 2 SEMESTER 1

- Photography in Production
- Pengajian Islam Kontemporari /
- Ethics and Moral 3 (Pelajar Bukan Muslim)
- 3D Character Modeling and Texturing
- Digital Storyboard
- Basic Animation
- Computing and Multimedia Project for Community

YEAR 2 SEMESTER 2

- 3D Animation Acting and Body 1
- Human Computer Interaction
- Intermediate Arabic / Intermediate Mandarin / Intermediate French

YEAR 2 SEMESTER 3

- Lighting & Rendering
- 3D Animation Acting and Body 2
- Editing for Production
- Production Management
- Final Year Project 01

YEAR 3 SEMESTER 1

- Compositing for Production
- Digital Entrepreneurship
- Videography in Production
- Kursus Integriti dan Antirasuah (KIAR)
- Final Year Project 02

YEAR 3 SEMESTER 2

- Kemahiran Kesantunan Berbahasa
- Web Design Production
- Branding and Advertising in Graphic Design / Introduction to Augmented Reality

YEAR 3 SEMESTER 3

- Industrial Training

LIST OF ELECTIVE COURSES

- Elective 1: Beginner Arabic / Beginner Mandarin / Beginner French
- Elective 2: Intermediate Arabic / Intermediate Mandarin / Intermediate French
- Elective 3: Branding and Advertising in Graphic Design / Introduction to Augmented Reality



"Animation can explain whatever the mind of man can conceive. This facility makes it the most versatile and explicit means of communication yet devised for quick mass appreciation."

- Walt Disney -

Study Path

BACHELOR of ARTS in 3D ANIMATION and DIGITAL MEDIA (HONOURS) (CM201)

Career Opportunities in:

- Employment in Government or Private Sector
- Animator,
- 3D Modeller,
- Stop Motion Animator,
- Character Animator,
- Concept Artist,
- Storyboard Artist,
- Lighting Artist,
- Background Painter and Graphic Designer

Doctor of Philosophy in Arts And Design
and
PhD in Related Fields

Master of Arts in Fields such as Visual
Communication / New Media / Design
Technology
or
Master in Art And Design

Bachelor of Arts in 3D Animation and Digital Media
(Honours)

Foundation / Diploma / STPM / APEL / A-Level related
certification recognised by MQA



BACHELOR IN COMPUTER SCIENCE (HONOURS) (CC203)

The Bachelor of Computer Science (Honours) is a locally developed degree that blends theoretical knowledge with practical skills. It integrates mathematics, programming, engineering, and linguistics to prepare students for critical fields like Artificial Intelligence, Data Analytic Education and Digital Governance. As a scientific and practical discipline, the program focuses on algorithms, data structures, and key subfields including computer graphics, cybersecurity, and software development.

Students gain high employability through hands-on experience with modern software tools, fostering problem-solving and logical reasoning. This single-major structure aligns with industry demands, enabling graduates to solve complex problems in the fast-paced tech sector.

Graduates are prepared for diverse roles such as Software Engineer, Full-stack Developer, Systems Analyst, or UI/UX Designer in both public and private sectors. Additionally, the program provides a strong foundation for master's level studies in specialized fields like Software Engineering, Cybersecurity, or Information Science.

PROFESSIONAL CERTIFICATION



CompTIA

PROGRAMME REQUIREMENT, STRUCTURE & STUDY PATH



Programme Code	: CC203
MQA	: (N/0613/6/0104) (10/2029) (MQA/PA17736)
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Computing (Third Edition 2023)

ENTRY REQUIREMENT

- A pass in STPM with a minimum grade of C (CGPA 2.00) in any TWO (2) subjects; **OR**
- A pass in STAM with a minimum grade of Jayyid in any TWO (2) subjects; **OR**
- A pass in Matriculation or Foundation studies with a minimum CGPA of 2.00; **OR**
- Any other Diploma in Science and Technology (Level 4, MQF) with a minimum CGPA of 2.75. Candidates with a CGPA below 2.75 but more than 2.50 may be admitted subject to a thorough internal evaluation process;

AND a credit in:

- Additional Mathematics at SPM level or its equivalent; **OR**
- Candidates with a pass in Mathematics and any one science, technology or engineering subject at the SPM level or its equivalent. Candidates are required to take and pass a mathematics reinforcement course equivalent to Additional Mathematics at the SPM level. This subject must be offered in Semester 1 prior to enrolment and is to be offered as an unconditional offer.
- Any other Diploma in Science and Technology or Business Studies with a minimum CGPA of 2.5 may be admitted, subject to a rigorous internal assessment process and a credit in Mathematics at SPM level or its equivalent; **OR**
- Diploma Kemahiran Malaysia (DKM) / Diploma Vokasional Malaysia (DVM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval; **OR**
- Diploma Lanjutan Kemahiran Malaysia (DLKM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval.

ENGLISH COMPETENCY REQUIREMENT (INTERNATIONAL STUDENT)

For international students, a minimum score of Malaysian University English Test (MUET) Band 3.5 or Common European Framework of Reference for Languages (CEFR) High B1, or a score of 40 in the Test of English as a Foreign Language (TOEFL), or a score of 5.0 in the International English Language Testing System (IELTS), or 47 in the Pearson Test of English (PTE), or 107 in ELS, or 154 in Cambridge English is required. The validity period for TOEFL and IELTS is two years from the date of the test result.

Programme Structure

BACHELOR IN COMPUTER SCIENCE (HONOURS) (CC203)

YEAR 1 SEMESTER 1

- Programming Fundamentals
- Computer Organization and Architecture
- Calculus and algebra
- Statistics and Probability
- English For Professional Interaction
- Kemahiran Kesantunan Berbahasa

YEAR 1 SEMESTER 2

- English for Advanced Learner
- Mandarin for Communication
- Arabic for Communication
- French for Communication

YEAR 1 SEMESTER 3

- Advanced Programming
- Fundamental of Database
- Operating Systems
- Data Communication and Networking
- Discrete Mathematics
- Digital Entrepreneurship

YEAR 2 SEMESTER 1

- Algorithmic Data Structure
- System Analysis and Design
- System Administration
- Computer Systems and Networking
- Security in Networking
- Programming Paradigm
- Network Administration
- Academic Writing

YEAR 2 SEMESTER 2

- Khidmat Masyarakat 2
- Kursus Integriti dan Antirasuah
- Penghayatan Etika dan Peradaban
- Hubungan Etnik
- Falsafah dan Isu Semasa
- Web Application Development

YEAR 2 SEMESTER 3

- Design and Analysis of Algorithms
- Human Computer Interaction and Web API
- Software Project Management
- Database Management
- Routing and Switching Essentials
- Ethical Hacking and Penetration Testing
- Mobile Apps Development

YEAR 3 SEMESTER 1

- Intelligent System
- Computing Final Year Project 1
- Enterprise Networking, Security and Automation
- Digital Forensic
- Enterprise Software Development
- Network Programming
- Cybersecurity Operations
- Software Testing for Quality Assurance
- Social Issues and Professional Practice
- Computing and Multimedia Project for Community

YEAR 3 SEMESTER 2

- Ethics and Moral 3
- Pengajian Islam Kontemporari
- Bahasa Kebangsaan A
- Computing Final Year Project 2

YEAR 3 SEMESTER 3

- Industrial Training





*"Programming is not a science of code, it's a science of thought.
The computer program is a beautiful mechanism made out of pure logic."
- Grace Hopper -*

Study Path

BACHELOR IN COMPUTER SCIENCE (HONOURS) (CC203)

Career Opportunities in:

- Employment in Government or Private Sector
- Software Developer / Engineer
- Web / Apps Developer
- AI / ML Developer
- IT Consultant
- IT Support Specialist
- Database Administrator
- Cybersecurity Analyst
- Software Quality
- Assurance (QA) Analyst
- DevOps / Cloud Engineer
- UI / UX Designer

Doctor of Philosophy in Computer Science
and
PhD in Related Fields

Master of Science in Fields such as
Computer Science / Software Engineering /
Information Science / Information
Technology

Bachelor in Computer Science
(Honours)

Foundation / Diploma / STPM / APEL / A-Level related
certification recognised by MQA



THE BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN COMPUTER APPLICATION DEVELOPMENT (CT204)

The Bachelor of Information Technology (Honours) in Computer Application Development provides theoretical and practical expertise in software engineering and IT. Launched in 2020 and MQA-accredited since 2022, the program covers advanced programming, mobile and enterprise development, software testing, and data analytics.

Students learn to design and integrate applications using modern technologies through combined theoretical and hands-on activities. The curriculum emphasizes problem-solving and decision-making skills within the fast-growing IT sector. Graduates are prepared for private and public sector careers as software engineers, full-stack developers, system analysts, or UI/UX designers. Additionally, the degree serves as a foundation for master's level studies in fields such as Computer Science, Information Science, or Software Engineering.

PROFESSIONAL CERTIFICATION



CompTIA



Red Hat
Academy

PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CT204
MQA	: (R/213/6/0312) (02/2028) (MQA/FA9074)
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Computing (Third Edition 2024)

ENTRY REQUIREMENT

- Pass STPM with a minimum Grade C (GP 2.00) in any TWO (2) subjects; **OR**
- Pass STAM with a minimum Grade of Jayyid in any TWO (2) subjects; **OR**
- Pass Matriculation or Foundation studies with a minimum CGPA of 2.00; **OR**
- Diploma in non-computing with a minimum CGPA of 2.75. Candidates with a CGPA below 2.75 but more than 2.50 can be admitted subject to a thorough rigorous assessment; **AND**
- credit in: Mathematics at SPM level or its equivalent; **OR**
- Candidates with a pass in Mathematics at SPM level need to take and pass the reinforcement Mathematics subject that is equivalent to the SPM level. The reinforcement Mathematics subject must be offered in first semester or before enrolment with unconditional offer. **OR**
- Diploma in Computing fields or equivalent with a minimum CGPA of 2.50. Candidates with a CGPA below 2.50 but more than 2.00 may be admitted subject to a thorough rigorous assessment; **OR**
- Pass DKM / DVM in Computing fields with a minimum CGPA of 2.50 **OR**
- Pass DLKM in Computing fields with a minimum CGPA of 2.50; **OR**
- Pass A-Level with minimum Grade D (GP 2.00) in any TWO (2) subjects including Mathematics; **OR**
- Pass UEC with minimum Grade B in any FIVE (5) subjects including Mathematics; **OR**
- Pass International Baccalaureate (IB) with at least 24 points; **OR**
- Other relevant and equivalent qualifications recognised by the Malaysian Government.

ENGLISH COMPETENCY REQUIREMENT (INTERNATIONAL STUDENT)

For international students, a minimum score of Malaysian University English Test (MUET) Band 3.5 or Common European Framework of Reference for Languages (CEFR) High B1, or a score of 40 in the Test of English as a Foreign Language (TOEFL), or a score of 5.0 in the International English Language Testing System (IELTS), or 47 in the Pearson Test of English (PTE), or 107 in ELS, or 154 in Cambridge English is required. The validity period for TOEFL and IELTS is two years from the date of the test result.

Programme Structure

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN
COMPUTER APPLICATION DEVELOPMENT (CT204)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Fundamental of Database
- Statistics and Probability
- Programming Fundamentals
- Academic Writing
- Arabic for Communication
- Mandarin for Communication
- French for Communication

YEAR 1 SEMESTER 2

- Operating Systems
- Software Development Paradigms
- Discrete Mathematics
- Algorithmic Data Structures
- Digital Entrepreneurship
- Penghayatan Etika dan Peradaban/
Bahasa Melayu Komunikasi 2
(International Student)

YEAR 1 SEMESTER 3

- Data Communication and Networking
- System Analysis and Design
- Cybersecurity Principles and Ethics

YEAR 2 SEMESTER 1

- Database Management
- Computer Systems and Networking
- Web Application Development
- Platform Technologies
- User Experience Design
- Falsafah dan Isu Semasa

YEAR 2 SEMESTER 2

- Software Project Management
- Advanced Application Development
- Mobile Apps Development
- IoT Technologies and Applications
- Computing and Multimedia Project
For Community
- Web API Development

YEAR 2 SEMESTER 3

- Software Design
- Python Programming Language
- Bahasa Kebangsaan A or
Pengajian Islam Kontemporari or
Ethics and Moral 3

YEAR 3 SEMESTER 1

- Kursus Integriti dan Antirasuah (KIAR)
- Final Year Project 1
- Enterprise Software Development
- Introduction to Data Science
- Software Testing and Quality Assurance
- Artificial Intelligence

YEAR 3 SEMESTER 2

- Industrial Training

YEAR 3 SEMESTER 3

- Final Year Project 2
- Cloud Computing

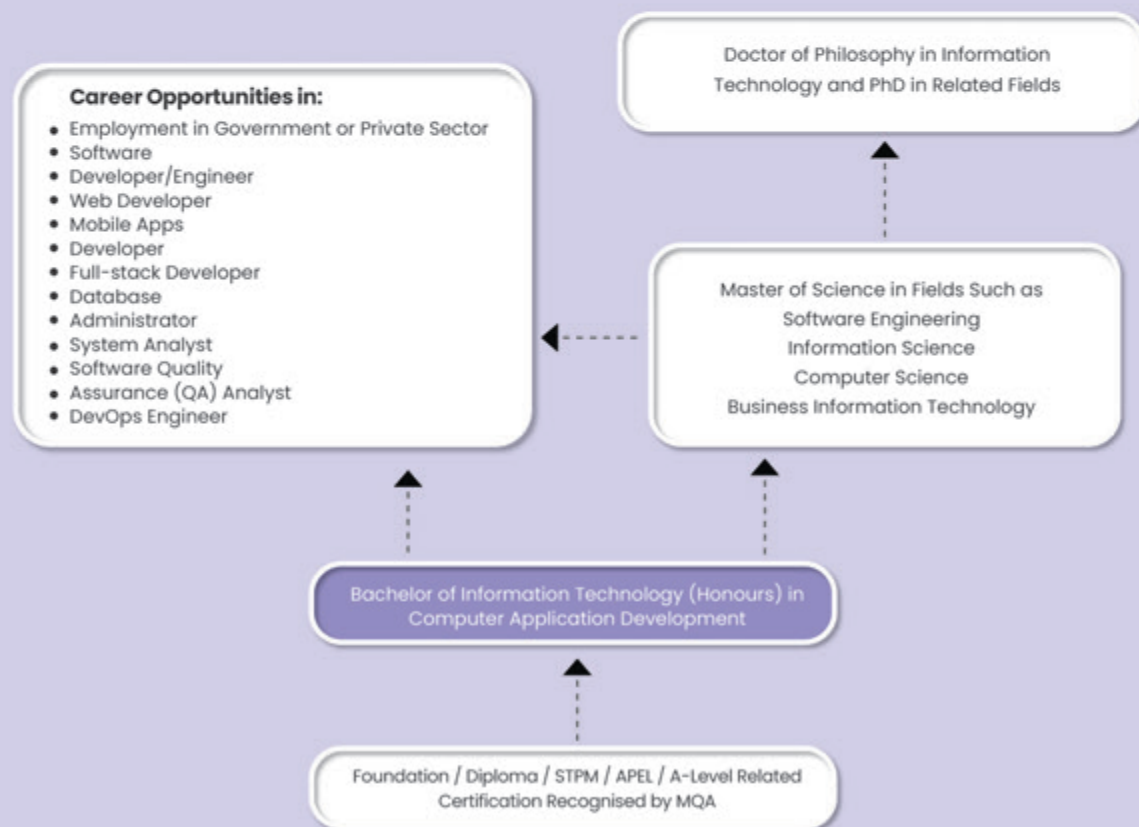




*"Design is not just what it looks like and feels like.
Design is how it works."
- Steve Jobs -*

Study Path

BACHELOR INFORMATION TECHNOLOGY (HONOURS)
IN COMPUTER APPLICATION DEVELOPMENT (CT204)



BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN CYBER SECURITY (CT206)

The Bachelor of Information Technology (Honours) in Cyber Security is a homegrown program providing theoretical and practical expertise for the modern workplace. Launched in 2015 and fully MQA-accredited since 2018, the curriculum covers key areas like cryptography, ethical hacking, digital forensics, and network security. Students gain experience designing and managing security infrastructure through a blend of academic study and hands-on training.

Upon completion, graduates possess the technical skills to solve complex problems and make informed decisions in the evolving cybersecurity landscape. They are prepared for diverse roles, including security engineer, analyst, consultant, and IT auditor, within both public and private sectors. Additionally, the degree enables students to pursue master's level studies in specialized fields such as Computer Security, Software Engineering, or Information Science.

PROFESSIONAL CERTIFICATION



CompTIA



Red Hat
Academy



PROGRAMME REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CT206
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Computing (3rd Edition 2024)

ENTRY REQUIREMENT

- A pass in STPM with a minimum grade of C (CGPA 2.00) in any TWO (2) subjects; **OR**
- A pass in STAM with a minimum grade of Jajyyid in any TWO (2) subjects; **OR**
- A pass in Matriculation or Foundation studies with a minimum CGPA of 2.00;
- Diploma (Level 4, MQF) with a minimum CGPA of 2.75. Candidates with a CGPA below 2.75 but more than 2.50 may be admitted subject to a thorough rigorous assessment;

AND a credit in:

- Mathematics at SPM level or its equivalent; **OR**
- Candidates with a pass in Mathematics at SPM level need to take and pass the reinforcement Mathematics subject that is equivalent to the SPM level. The reinforcement Mathematics subject must be offered in the first semester or before enrolment with unconditional offer. **OR**
- Diploma in Computing (Level 4, MQF) or equivalent with a minimum CGPA of 2.50. Candidates with a CGPA below 2.50 but more than 2.00 may be admitted subject to a thorough rigorous assessment; **OR**
- Diploma Kemahiran Malaysia (DKM) / Diploma Vokasional Malaysia (DVM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval; **OR**
- Diploma Lanjutan Kemahiran Malaysia (DLKM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval; **OR**
- Other relevant and equivalent qualifications recognised by Malaysian Government. (Candidate can be admitted if their admission qualification contains Mathematics subject(s) equivalent to Mathematics at the SPM level. If it is not equivalent, the reinforcement Mathematics subject equivalent to the SPM level must be offered in the first semester or before enrolment with unconditional offer).

ENGLISH COMPETENCY REQUIREMENT (INTERNATIONAL STUDENT)

Achieve a minimum score of 5.0 in the IELTS or equivalent. If a student does not meet this requirement, the HEP must offer English proficiency courses to ensure that the student's proficiency is sufficient to meet the needs of the programme.

- Students are required to pass the reinforcement Mathematics before being allowed to take related core courses. The candidate can sit for any subject that did not indicate Mathematics as a prerequisite.
- Reinforcement Mathematics can contribute to the overall graduating credit.
- Students from Matriculation / Foundation or its equivalent can be exempted from taking reinforcement Mathematics, provide that the Mathematics offered at the programme level is equivalent / more than the Additional Mathematics offered at an SPM level.

Programme Structure

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
IN CYBER SECURITY (CT206)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Academic Writing
- Data Communication and Networking
- Statistics and Probability
- Programming Fundamentals
- Mandarin for Communication /
Arabic for Communication /
French for Communication

YEAR 2 SEMESTER 1

- Platform Technologies
- Database Management
- Routing and Switching Essentials
- Fundamental of Cryptography
- Web Application Development
- System Analysis and Design

YEAR 3 SEMESTER 1

- Final Year Project 1
- Pengajian Islam Kontemporari /
Ethics and Moral 3 / Bahasa Kebangsaan A
- Kursus Integriti dan Antirasuah
- Cybersecurity Operations
- Digital Forensic
- Enterprise Networking, Security and Automation

YEAR 1 SEMESTER 2

- Operating Systems
- Fundamental of Database
- Discrete Mathematics
- Computer Systems and Networking
- Algorithmic Data Structures
- Digital Entrepreneurship

YEAR 2 SEMESTER 2

- Software Project Management
- Ethical Hacking and Penetration Testing
- Security in Networking
- Web API Development
- Mobile Apps Development
- Computing and Multimedia Project For
Community

YEAR 3 SEMESTER 2

- Industrial Training

YEAR 1 SEMESTER 3

- User Experience Design
- Cybersecurity Principles and Ethics

YEAR 2 SEMESTER 3

- Cloud Computing
- Penghayatan Etika dan Peradaban
- Falsafah dan Isu Semasa /
Bahasa Melayu Komunikasi 2
(International Student)

YEAR 3 SEMESTER 3

- Final Year Project 2
- Information Security Assurance
Artificial Intelligence

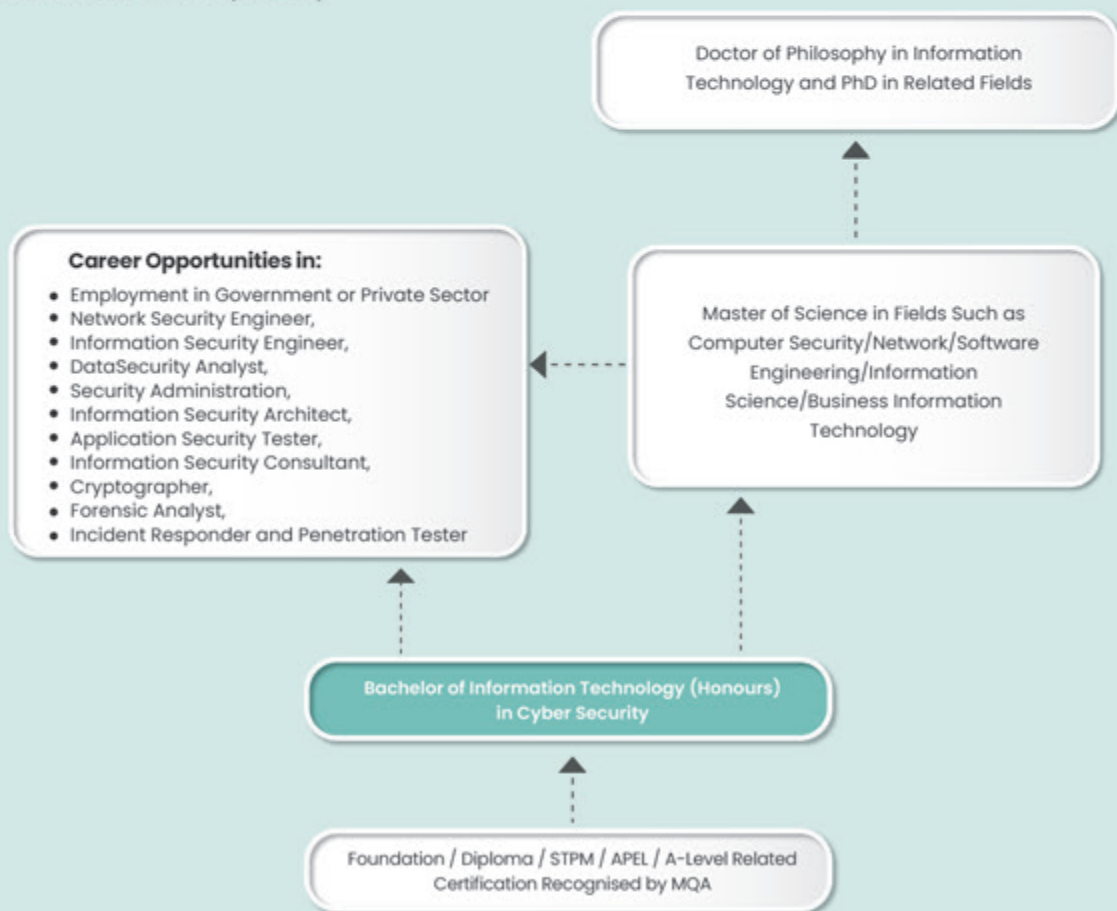




"As the world becomes more interconnected, computing and networks are no longer just tools; they are the infrastructure of our daily lives. Defending them is an ongoing act of vigilance."
- Jeannette Wing -

Study Path

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
IN CYBER SECURITY (CT206)



BACHELOR IN INFORMATION TECHNOLOGY (HONOURS) (DATA ANALYTICS) (CT207)

The Bachelor in Information Technology (Honours) in Data Analytics is a specialized program designed to provide students with a solid foundation in both information technology and data analytics. It combines comprehensive theoretical knowledge with hands-on practical skills, enabling graduates to meet the demands of today's data-driven economy. This program is a rebranded version of the Bachelor of Information Technology (Honours) in Business Computing, which began its first intake in May 2018. The original program received provisional accreditation from the Malaysian Qualifications Agency (MQA) in 2017, and was subsequently granted full accreditation in 2021.

Students will be exposed to core IT subjects such as programming, database management, networking, and information systems, alongside advanced data analytics modules including statistical analysis, machine learning, big data technologies, and data visualization. Through industry-relevant projects, case studies, and internships, students will gain real-world experience in transforming raw data into actionable insights that drive strategic decision-making. This program emphasizes critical thinking, problem-solving, and analytical skills, while also fostering ethical and responsible use of data. By the time they graduate, students will be equipped to work effectively with modern analytics tools and technologies, ensuring they remain competitive in an ever-evolving digital landscape. Upon graduation, students will possess the expertise to apply advanced data analytics techniques in extracting actionable insights and optimizing decision-making across various business domains. As data becomes the backbone of modern enterprises, professionals with strong analytical capabilities are increasingly in demand.

Graduates of this program will be well-prepared for careers such as Data Analyst, Business Intelligence Specialist, Data Scientist, Data Engineer, and Consultant, serving industries that include finance, healthcare, retail, technology, and government. The program's emphasis on analytical thinking, problem-solving, and data-driven project management ensures that graduates not only excel in technical roles but also hold a competitive advantage in a wide range of positions within today's technology-driven economy.

PROFESSIONAL CERTIFICATION



CompTIA



Red Hat
Academy



PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CT207
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English / Bahasa Melayu (MPU)
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Computing (Third Edition 2024)

ENTRY REQUIREMENT

- A pass in STPM with a minimum grade of C (CGPA 2.00) in any TWO (2) subjects; **OR**
- A pass in STAM with a minimum grade of Jayyid in any TWO (2) subjects; **OR**
- A pass in Matriculation or Foundation studies with a minimum CGPA of 2.00;
- Diploma (Level 4, MQF) with a minimum CGPA of 2.75. Candidates with a CGPA below 2.75 but more than 2.50 may be admitted subject to a thorough rigorous assessment;
- Mathematics at SPM level or its equivalent; **OR**
- Candidates with a pass in Mathematics at SPM level need to take and pass the reinforcement Mathematics subject that is equivalent to the SPM level. The reinforcement Mathematics subject must be offered in the first semester or before enrolment with unconditional offer. **OR**
- Diploma in Computing (Level 4, MQF) or equivalent with a minimum CGPA of 2.50. Candidates with a CGPA below 2.50 but more than 2.00 may be admitted subject to a thorough rigorous assessment; **OR**
- Diploma Kemahiran Malaysia (DKM) / Diploma Vokasional Malaysia (DVM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval; **OR**
- Diploma Lanjutan Kemahiran Malaysia (DLKM) in Computing fields with a minimum CGPA of 2.50 subjected to HEP Senate / Academic Board's approval; **OR**
- Other relevant and equivalent qualifications recognised by Malaysian Government. (Candidate can be admitted if their admission qualification contains Mathematics subject(s) equivalent to Mathematics at the SPM level. If it is not equivalent, the reinforcement Mathematics subject equivalent to the SPM level must be offered in the first semester or before enrolment with unconditional offer).

ENGLISH COMPETENCY REQUIREMENT (INTERNATIONAL STUDENT)

Achieve a minimum score of 5.0 in the IELTS or equivalent. If a student does not meet this requirement, the HEP must offer English proficiency courses to ensure that the student's proficiency is sufficient to meet the needs of the programme.

- Students are required to pass the reinforcement Mathematics before being allowed to take related core courses. The candidate can sit for any subject that did not indicate Mathematics as a prerequisite.
- Reinforcement Mathematics can contribute to the overall graduating credit.
- Students from Matriculation / Foundation or its equivalent can be exempted from taking reinforcement Mathematics, provide that the Mathematics offered at the programme level is equivalent / more than the Additional Mathematics offered at an SPM level.

Programme Structure

BACHELOR IN INFORMATION TECHNOLOGY
(HONOURS) (DATA ANALYTICS) (CT207)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Fundamental of Database
- Statistics and Probability
- Programming Fundamentals
- Academic Writing
- Arabic for Communication /
Mandarin for Communication /
French for Communication

YEAR 1 SEMESTER 2

- Operating Systems
- Programming Language For Data
Science
- Discrete Mathematics
- Algorithmic Data Structures
- Digital Entrepreneurship
- Penghayatan Etika dan Peradaban/
Bahasa Melayu Komunikasi 2
(International Student)

YEAR 1 SEMESTER 3

- Data Communication and Networking
- System Analysis and Design
- Cybersecurity Principles and Ethics

YEAR 2 SEMESTER 1

- Database Management
- Computer Systems and Networking
- Web Application Development
- Platform Technologies
- User Experience Design
- Falsafah dan Isu Semasa

YEAR 2 SEMESTER 2

- Software Project Management
- E-Commerce
- Mobile Apps Development
- Introduction to Data Mining
- Computing and Multimedia Project
For Community
- Web API Development

YEAR 2 SEMESTER 3

- Introduction to Data Visualization
- Business Intelligence
- Bahasa Kebangsaan A
Pengajian Islam Kontemporari
Ethics and Moral 3

YEAR 3 SEMESTER 1

- Kursus Integriti dan Antirasuah (KIAR)
- Final Year Project 1
- Advance Data Mining with Machine Learning
- Big Data Technologies
- Management Information System
- Artificial Intelligence

YEAR 3 SEMESTER 2

- Industrial Training

YEAR 3 SEMESTER 3

- Final Year Project 2
Cloud Computing





"Information is the oil of the 21st century,
and analytics is the combustion engine."

-Peter Sondergaard-

Study Path

BACHELOR IN INFORMATION TECHNOLOGY
(HONOURS) (DATA ANALYTICS) (CT207)

Career Opportunities in:

- Employment in Government Or Private Sector
- Business Analyst
- Data Analyst
- Data Scientist
- Data Engineer
- Business Consultant
- Business Software Application Developers
- Software Tester
- Database Developer
- System Analyst
- Database Administrators
- Internet Applications Developers
- Product Support Or In It Sales Support/ Consultant

Doctor of Philosophy in Information
Technology and PhD in Related Fields

Master of Science in Fields such as
Computer Science / Information
System / Business Information
Technology
or
Master of Business Administration

Bachelor of Information Technology (Hons) in
Data Analytics

Foundation / Diploma / STPM / APEL / A-Level related
certification recognised by MQA



BACHELOR IN ARTIFICIAL INTELLIGENCE (HONOURS) (CC204)

The Bachelor in Artificial Intelligence (Honours) is a specialised undergraduate programme developed in line with the Programme Standards: Artificial Intelligence (PS:AI 2025) by the Malaysian Qualifications Agency (MQA). The programme provides students with strong theoretical and practical knowledge in Artificial Intelligence (AI), focusing on the design, development, and management of intelligent digital solutions. The curriculum covers key areas such as AI fundamentals, machine learning, generative AI, reinforcement learning, analytics, entrepreneurship, Final Year Project, and industrial training. Teaching and learning activities include lectures, laboratory sessions, project-based learning, and industry engagement to enhance technical and professional competencies.

Graduates are equipped with knowledge and skills to develop AI-based solutions for real-world challenges while applying ethical and responsible AI practices. Career opportunities include roles such as Artificial Intelligence Engineer, Machine Learning Engineer, Data Scientist, and AI Systems Developer across industries including finance, healthcare, manufacturing, cybersecurity, and government technology. The programme also provides a strong foundation for postgraduate studies in Artificial Intelligence and related fields.

PROFESSIONAL CERTIFICATION

CompTIA

PROGRAMME

REQUIREMENT, STRUCTURE & STUDY PATH

Programme Code	: CC204
MQA	: (N/0613/6/0146) (11/30) (MQA/PA18486)
Duration	: 3 years
Total Credit Hours	: 120
Medium of Instruction	: English
Awarding Body	: Universiti Poly-Tech Malaysia
Programme Standard	: Artificial Intelligence (2025 Edition)

ENTRY REQUIREMENT

- A pass in Sijil Tinggi Persekolahan (STPM) (Arts Stream) with a minimum Grade of C (GP 2.00) in any TWO (2) subjects; **OR**
- A pass in Sijil Tinggi Agama Malaysia (STAM) with a minimum Grade of Jayyid in any TWO (2) subjects; **OR**
- A pass in Matriculation or Foundation studies with a minimum CGPA of 2.00; **OR**
- Any Diploma in Science and Technology (Level 4 Malaysian Qualifications Framework, MQF) with a minimum CGPA of 2.75. Candidates with a CGPA below 2.75 but more than 2.50 may be admitted subject to a thorough rigorous assessment; **OR**

AND a credit in:

- Additional Mathematics at the Sijil Pelajaran Malaysia (SPM) level or its equivalent; **OR**

Mathematics and any one of the Science, Technology or Engineering subjects at the SPM level or its equivalent. Candidates need to take and pass the reinforcement Mathematics equivalent to Additional Mathematics at the SPM level. This subject must be offered in the first semester or before enrolment with an unconditional offer; **OR**

A pass in STPM (Science Stream) or its equivalent with a minimum Grade of C (GP2.00) in Mathematics subject and ONE (1) Science / ICT subject; **OR**

Diploma in Computing fields (Level 4, MQF) or equivalent with a minimum CGPA of 2.50. Candidates with a CGPA below 2.50 but more than 2.00 may be admitted subject to a thorough rigorous assessment; **OR**

Diploma Kemahiran Malaysia (DKM) / Diploma Vokasional Malaysia (DVM) in Computing fields with a minimum CGPA of 2.5, subject to HEP Senate / Academic Board's approval; **OR**

Diploma Lanjutan Kemahiran Malaysia (DLKM) in Computing fields with a minimum CGPA of 2.50, subject to HEP Senate / Academic Board's approval.

Other relevant and equivalent qualifications recognised by the Malaysian Government. (Candidates can be admitted if their admission qualification contains Mathematics subject (s) equivalent to Additional Mathematics at the SPM level. If it is not equivalent, the reinforcement Mathematics subject that is equivalent to the SPM level must be offered in the first semester or before enrolment with an unconditional offer).

ENGLISH COMPETENCY REQUIREMENT (INTERNATIONAL STUDENT)

Achieve a minimum of Band 3.5 in Malaysian University English Test (MUET) or equivalent to Common European Framework of Reference for Languages (CEFR) (High B1). If a student does not meet this requirement, the HEP must offer English proficiency courses to ensure that the student's proficiency is sufficient to meet the needs of the programme.

Notes:

- Students are required to pass the reinforcement Mathematics before being allowed to take related core courses. The candidate can sit for any subject that did not indicate Mathematics as a prerequisite
- Reinforcement Mathematics can contribute to the overall graduating credit.
- Students from Matriculation / Foundation or its equivalent can be exempted from taking reinforcement Mathematics, provide that the Mathematic offered at the programme level is equivalent / more than the Additional Mathematics offered at an SPM level.
- For candidates who have not passed the rigorous internal assessment, the HEP can determine for the candidate to be assessed for entry to the programme to pass preparatory courses as determined by the HEP before entering the programme, and subject to the HEP Senate / Academic Board's approval.
- Rigorous assessment can be done through interviews, portfolios, written tests, or any form of assessment.
- Reinforcement Mathematics can be offered as part of or outside the curriculum structure. If Reinforcement Mathematics is offered as part of the curriculum structure, it should count as an addition to the total number of credits required for graduation.
- For Private Higher Educational Institutions: Refer to Surat JPT/GS 1000-606 Jld. 2(23), 21st April, 2020 - Kemasukan Pelajar Lulusan Diploma Kemahiran Malaysia (DKM) Diploma Lanjutan Kemahiran Malaysia (DLKM) dan Diploma Vokasional Malaysia (DVM) ke Peringkat Sarjana Muda (Tahap 6 MQF) atau yang setara dengannya di Institusi Pendidikan Tinggi Swasta.

Programme Structure

BACHELOR IN ARTIFICIAL INTELLIGENCE (HONOURS) (CC204)

YEAR 1 SEMESTER 1

- Computer Organization and Architecture
- Programming Fundamentals
- Introduction to Artificial Intelligence
- Statistics and Probability
- English For Professional Interaction
- Kemahiran Kesantunan Berbahasa

YEAR 1 SEMESTER 2

- Falsafah dan Isu Semasa
- English for Advanced Learner
- Mandarin for Communication / Arabic for Communication / French for Communication

YEAR 1 SEMESTER 3

- Algorithmic Data Structure
- Fundamental Database
- Operating System
- Data Communication and Networking
- Discrete Mathematics
- Digital Entrepreneurship

YEAR 2 SEMESTER 1

- Machine Learning
- Agents and Cognitive Science
- Responsible AI and ethics
- Natural Language Processing
- Data Intelligence
- Academic Writing

YEAR 2 SEMESTER 2

- Kursus Integriti dan Antirasuah
- Deep Learning
- Generative AI

YEAR 2 SEMESTER 3

- Reinforcement Learning
- Perception and Computer Vision
- Penghayatan Etika dan Peradaban
- NextGen System Development
- Big Data Technologies
- Digital Transformation

YEAR 3 SEMESTER 1

- Final Year Project 1
- Social Media Analytics
- AI for Business Strategy
- Semantic Technology
- AI in Supply Chain
- Image Analytic
- AI in Finance
- Business Intelligence
- Computing and Multimedia Project for Community

YEAR 3 SEMESTER 2

- Ethics and Moral
- Pengajian Islam Kontemporari
- Bahasa Kebangsaan A
- FYP 2

YEAR 3 SEMESTER 3

- Industrial Training



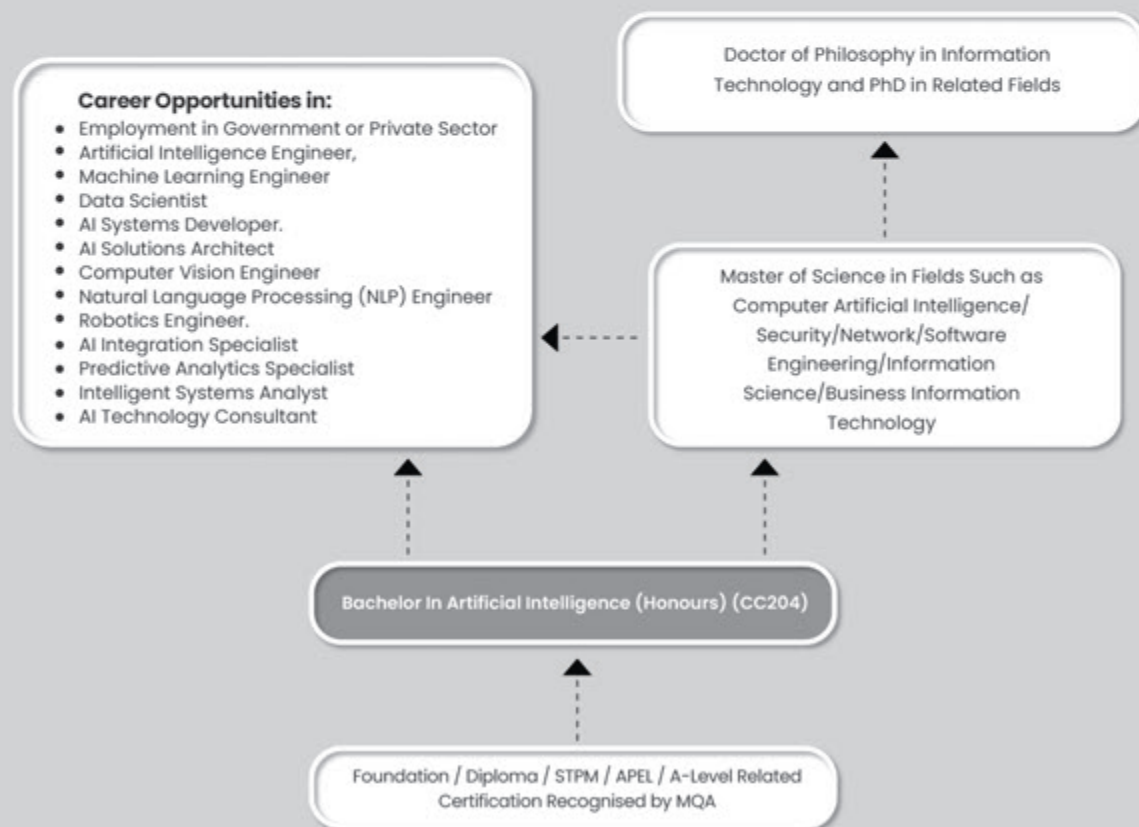


"The question is not whether machines think but whether men do. We are equipping intelligent systems not to replace us, but to amplify our potential to solve the unsolvable."

-B.F. Skinner-

Study Path

BACHELOR IN ARTIFICIAL INTELLIGENCE (HONOURS) (CC204)





CISCO CERTIFIED: LESS THEORY, MORE REALITY

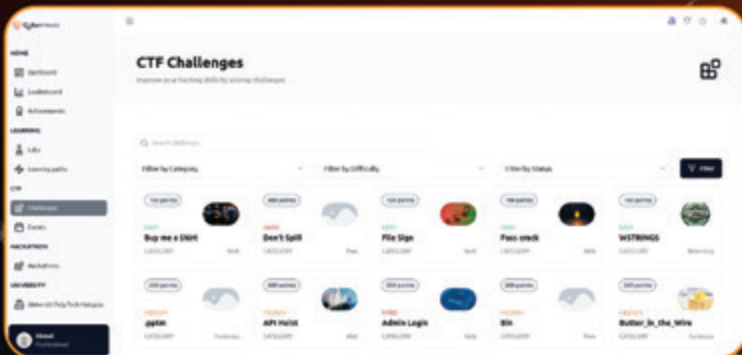
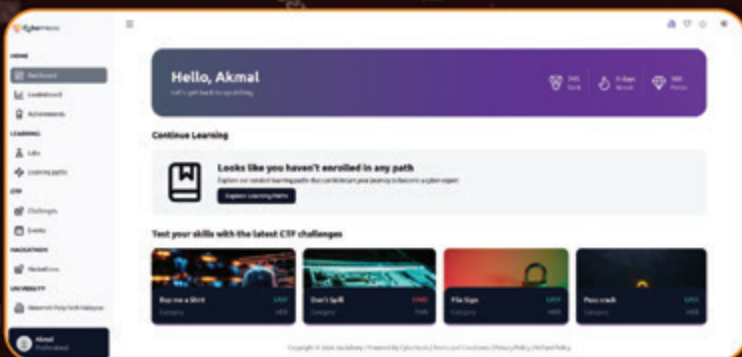
Through the CCNA 1, CCNA 2 & CCNA 3 courses embedded in our academic programmes, students are exposed to a comprehensive and industry-aligned learning experience in computer networking. These courses combine fundamental networking concepts with hands-on practical activities, enabling students to develop essential skills in network configuration, switching and routing, IP addressing, wireless networking, security fundamentals, and enterprise network management.

This initiative reflects our commitment to producing graduates who are technically competent, certification-ready, and prepared to meet the evolving demands of the digital and networking industry.



What is Hackdemy?

Hackdemy is a next-gen cyber range platform delivering immersive, real-world cybersecurity training designed to close the skill gap. At Hackdemy, we're not just educating we're building a resilient future.



hackdemy



Why Hackemy?

- Highly Experienced Team
- Tailored Training Programs For Schools, Colleges And Working Professionals
- Customized Job Role Specific Learning Paths
- Cyber Range Lab Platform for Hands-On Training
- Affordable LMS Platform and Accessible Anywhere

**It's time to
level up your skills
with real-world training**



What Hackemy Does?

- Virtual Cloud Based Cyber Range
- Conducting Hackathon / CTF Challenges
- Helping Universities for Structured Learning Paths
- Interactive Gamified Learning Platform
- Realistic Simulations by Zero Day Researchers



Comprehensive Education Beyond Theoretical Knowledge

The university's committed educators work tirelessly to ensure that every student receives personalized attention and support that enables them to realize their full potential.....



Nurturing Professionals Impacting The Nation.



AINUR AZRA BINTI ABD GHANI

 **UMINI: (CM201) Bachelor of Arts in 3D Animation and Digital Media (Honours)**

(3D Layout Artist and Animator)
Monsta Studio Sdn Bhd

"My journey into the creative industry started at UPTM, where I studied 3D Animation. The hands-on projects and support from my lecturers helped me develop the skills and confidence I needed to grow as an artist and eventually step into a professional studio environment. Today, I'm grateful to be part of Monsta and contribute to animation projects that reach audiences both locally and internationally. It's been an exciting journey, and I'm thankful that UPTM helped lay the foundation for where I am today."



AMIRUL ARIF BIN SHAHROM

 **UMINI: (CM201) Bachelor of Arts in 3D Animation and Digital Media (Honours)**

(Games development lecturer)
Abingdon & Witney College, United Kingdom

The multidisciplinary approach of the 3D Animation and Digital Media program at UPTM was instrumental in shaping my trajectory as both a creative professional and an educator. The curriculum seamlessly bridges artistic execution with technical theory, instilling a deep understanding of the global creative ecosystem. I am deeply grateful to MARA for its financial support through study loans during both my Bachelor's and Master's degree journeys, which enabled me to pursue higher education and professional growth. The academic excellence and strategic insights fostered by the faculty gave me the confidence and capability to transition into international academia, where I now train the next generation of game developers in the United Kingdom.

Success Stories



ARIIQMAN NAUFAL BIN OTHMAN ZAILANI

Alumni: (CT204) The Bachelor of Information Technology (Honours) in Computer Application Development

(Data Scientist) AstraZeneca Malaysia

"My experience at AstraZeneca as a Data Scientist within Global Finance Services has given me the opportunity to collaborate and consult with various process towers across global sites, including Poland, Costa Rica, and the United Kingdom. My role involves understanding business needs, analyzing challenges, and proposing solutions that help improve operational efficiency and achieve organizational objectives. This experience has strengthened my analytical thinking, problem-solving, communication, and collaboration skills in a global working environment. I am grateful to the Faculty of Computer and Information Technology at University Poly-Tech Malaysia for providing the knowledge and foundation that have supported my professional growth and career development."



NURAZLYA NATASHA BINTI ABDULLAH

Alumni: (CT206) Bachelor of Information Technology (Honours) In Cyber Security

(CT301) Master of Science In Information Systems

(ITExecutive) IPSB Technology

"UPTM's Cyber Security and Information Systems programs equipped me with the advanced analytical and technical capabilities required to safeguard complex enterprise environments. Today, I apply these skills as an IT Executive at IPSB Technology, a leader in driving Malaysia's largest broadcasting systems, an authorized Apple reseller, and an AWS partner. The professional guidance and rigorous training I received at UPTM were foundational to my career, enabling me to confidently deliver secure and scalable technology solutions at an industry-leading level."



BE PART OF OUR AMAZING STORIES...



Times Higher Education
Impact Rankings

AppliedHE

Scan For Further Information



UNIVERSITI POLY-TECH MALAYSIA

Jalan 6/91 Taman Shamelin Perkasa

56100, Kuala Lumpur

MALAYSIA

Email : marketing@uptm.edu.my

Tel : +60392069700

Tel : +60392837186 (Marketing Department)

www.uptm.edu.my



uptm_official